

Beyond this meeting

A key aspect of CoFI is the flexibility provided for exploration and experimentation in inversion

Created examples and brought existing capabilities together

Making unique approaches accessible through the common interface and grow the set of examples

A FAIR example driven framework...

A conversation about CoFI over coffee is just the beginning...



Explore our examples

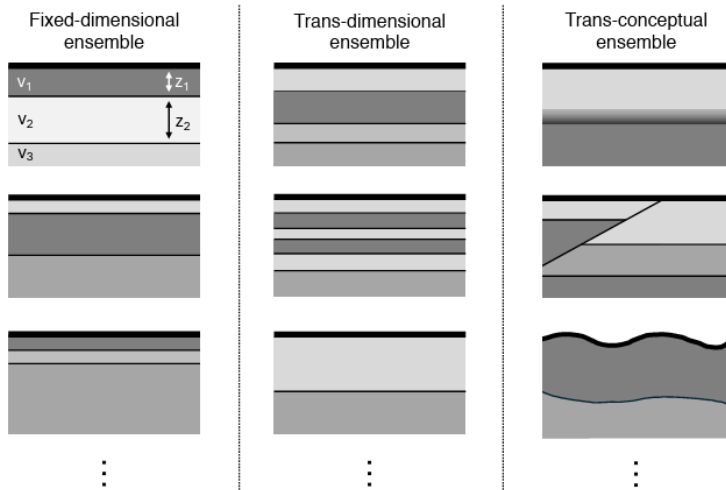
Setup your own inverse problem

Let us know what you think



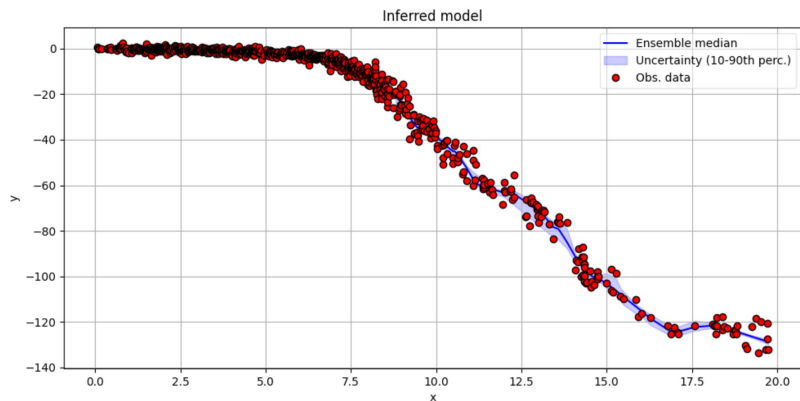
Upcoming capabilities...

Trans-conceptual sampling



<https://github.com/inlab-geo/pyTransC>

Options for trans-dimensional sampling



<https://github.com/fmagrini/bayes-bay>

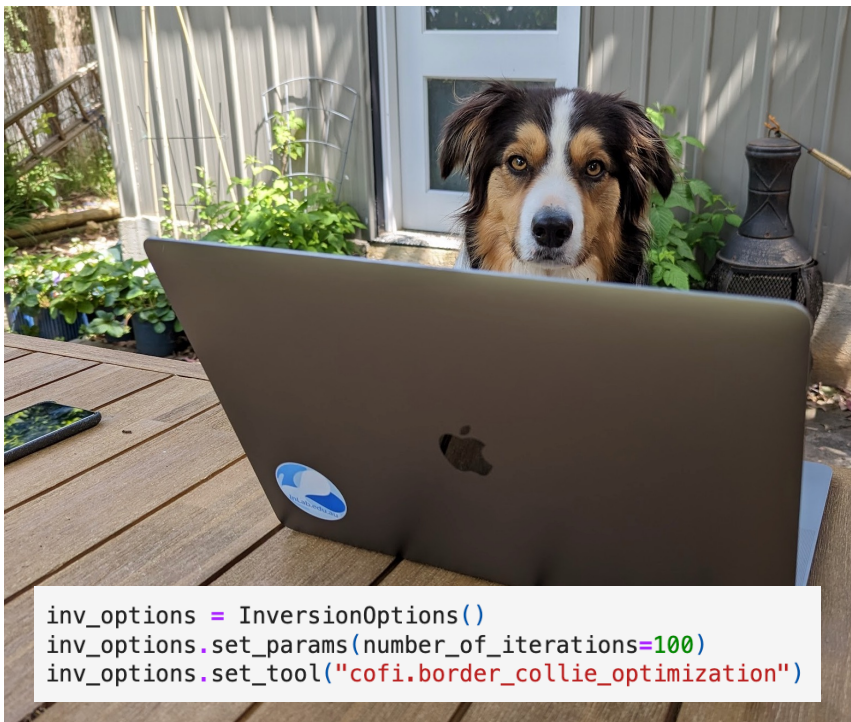
<https://github.com/inlab-geo/pyrjmcnc>

Unified Interface for ensemble methods

```
def log_probability(theta, x, y, yerr):  
    lp = log_prior(theta)  
    if not np.isfinite(lp):  
        return -np.inf  
    return lp + log_likelihood(theta, x, y, yerr)
```

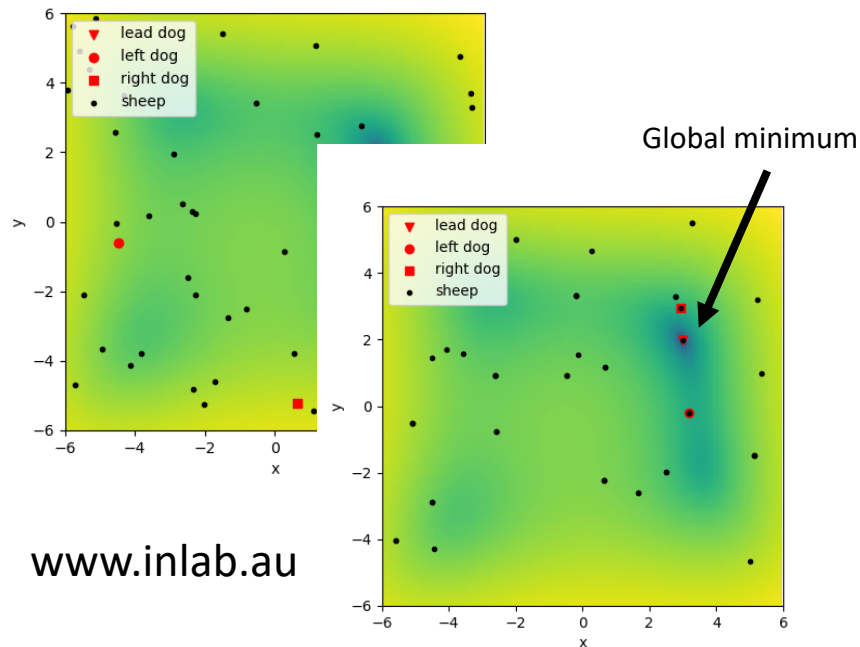
<https://emcee.readthedocs.io/en/stable/>

But what about my favourite metaheuristic optimiser?



```
inv_options = InversionOptions()  
inv_options.set_params(number_of_iterations=100)  
inv_options.set_tool("cofi.border_collie_optimization")
```

Border Collie Optimization



Wrap up

This is an **open source** project

Feedback:

Choosing priorities: Expanding the methods; the examples, building a community;

Wishlist includes: scalability (HPC); new examples; automatic differentiation, Surrogate modelling. Link to new optimization and sampling packages...

Work with us & get involved:

creating an application example using CoFI

join our slack community:

<http://inlab.au/community>

raise an issue on github:

<https://github.com/inlab-geo/cofi/issues>